

Aug. 2026

Hosted by Stanford, Coinbase, Localhost



A Workshop on Post-Quantum Bitcoin

Dan Boneh
Stanford University

'That Is The End Of Bitcoin'—The Quantum Race For \$470 Billion

By [Boaz Sobrado](#), Contributor. © Boaz Sobrado is a fintech leader and...



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Published Aug 02, 2026,

Jim Cramer Says He Plans to Sell All His Bitcoin on Quantum-Computing Fears

Published 11:03 AM Aug 3, 2026

Source

JH Kim

D-Wave CEO Warns Quantum Computing Could Eventually Break Bitcoin's Proof-of-Work

Bitcoin

Quantum Computing

Cryptocurrency Security

Proof Of Work

D-Wave

Jul 29, 2026 • 3 min read

by Dhaval

Bitcoin needs an answer ...

When quantum ??

This is the wrong question ...

Instead:

How to ensure Bitcoin survives beyond Q-day

... will restore present day trust in Bitcoin

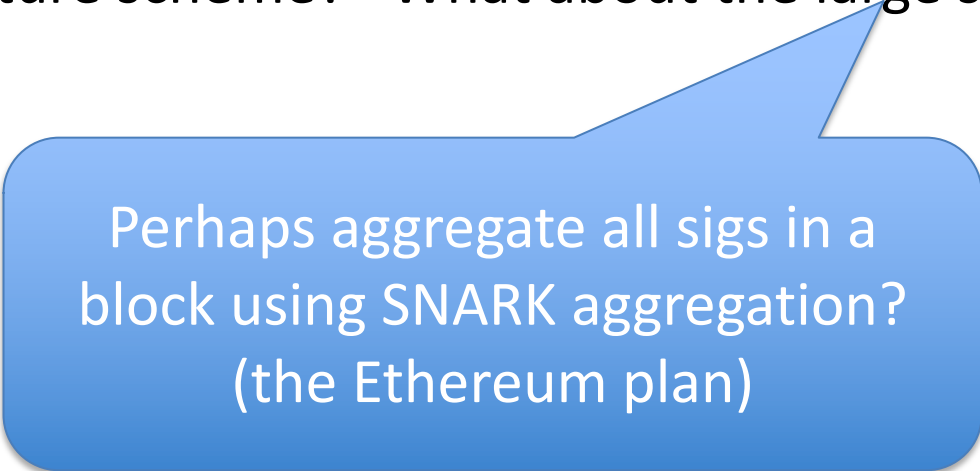
Today's workshop

- 9:30am – noon: three presentations
- noon: lunch
- 1pm: Panel: Output Script Type Design Considerations
- 2-4pm: four breakout sessions
 - Hash-based signatures: SHRINCS
 - Lattice signatures
 - Hardware wallets
 - Dropkick

Thank you JD for transcribing

Many questions to discuss

- What output types should Bitcoin holders transition to?
- Should we add an OP_CHECKPQSIG opcode?
 - What signature scheme? What about the large sig size?



Perhaps aggregate all sigs in a block using SNARK aggregation?
(the Ethereum plan)

Many questions to discuss

- What output types should Bitcoin holders transition to?
- Should we add an OP_CHECKPQSIG opcode?
 - What signature scheme? What about the large sig size?
- How to update hardware wallets? MPC wallets? tripwire?
- What happens the day Google announces they broke ECDSA?
 - More sinister: what happens if assets move on chain?
 - What if Q-day happens before Bitcoin has OP_CHECKPQSIG ?
- What is the recovery path for frozen assets?

Post-quantum Signature Families

- **Lattice-based signatures:** ML-DSA, Falcon, ~~HAWK~~
 my preference for blockchains
- **Hash-based signatures:** SLH-DSA, Shrincs, LeanSig, and others
- **Isogenies:** SQIsign
- **Multivariate:** MAYO, SNOVA, UOV, ...
- **ZK Proof of preimage:** Picnic, Feast, ...

Post-quantum Signature Families

- Lattice-based signatures: ML-DSA, Falcon, HAWK

- **Hash-based signatures:** SLH-DSA, Shrincs, LeanSig, and others

- Isogenies: SQIsign



currently the most likely option
(introduces no new assumptions)

- Multivariate: MAYO, SNOVA, UOV, ...

- ZK Proof of preimage: Picnic, Feast, ...

... but non-hardened
key derivation
seems impossible

Hash-based signatures: terminology

One-time signature (OTS): [Lamport, Winternitz (WOTS), WOTS+C]

- Secure if at most one signature is released for a public key
- Sig. len: **292 bytes** (WOTS+C: W=256, L=16) **516 bytes** (WOTS+C: W=16, L=32)
faster verify

Stateless Few-times signature (FTS): [HORS, FORS, FORS+C, PORS-FP]

- Secure if at most “few” sigs. are released for a public key
- Example sig. length: **2,448 bytes** (FORS+C: K=26, A=5)

security:

# sigs released	1	2	3	4	5
attack time	2^{128}	2^{104}	2^{89}	2^{78}	2^{70}

Hash-based signatures: terminology

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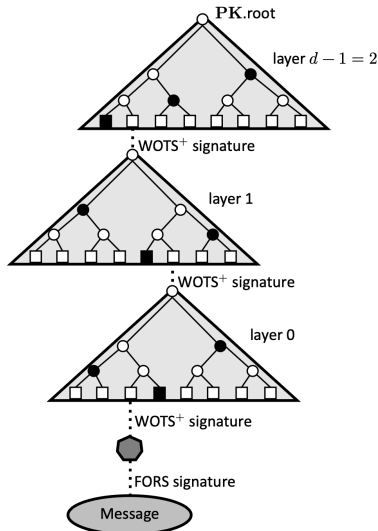
Stateful Few-times signature (SFTS): [XMSS, FXMSS, LMS]

- String together a few OTS public keys
- Example sig. length: **445 bytes** (XMSS: 2^{10} sigs)
- Signer must maintain state to ensure each pk is only used once
 - ⇒ a footgun?
 - ⇒ use hybrid sigs (SFTS and ECDSA) ??

NIST Hash-based signatures

NIST's **SLH-DSA** (based on SpHincS+)

Goal: designed for 2^{64} stateless signatures with one sk .



SLH-DSA-128: sig-size = **7.8 KB**

SLH-DSA-128f: sig-size = **17 KB**

(pk size = 32 bytes)

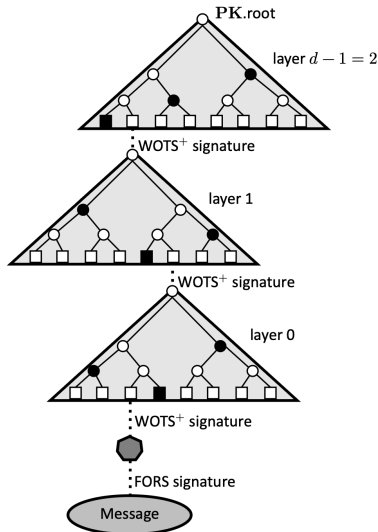
Well suited for a software update system
... poorly designed for blockchains

(FORS: a few times signature)

NIST Hash-based signatures

NIST's **SLH-DSA** (based on SpHincS+)

Goal: designed for 2^{64} stateless signatures with one sk .



SLH-DSA-128: sig-size = **7.8 KB**

SLH-DSA-128f: sig-size = **17 KB**

(pk size = 32 bytes)

More NIST params (Apr. 2026): **2^{24}** stateless sigs

SLH-DSA-SHA2-128-24: sig-size = **3.8 KB**

(FORS: a few times signature)

(sign time: $\approx 1B$ hashes, verify time: 311 hashes)

Sphincs-: more parameterizations

VARIANT C11 WOTS+C / FORS+C	d=2 h=16
SIGNATURE 3,976 B	
SIGN HASHES 292 K	

only 2^{14} sigs

VARIANT C12 vanilla SPHINCS+ (SPX)	d=5 h=20
SIGNATURE 6,512 B	
SIGN HASHES 36.6 K	

only 2^{14} sigs
fast sign, but long sig

VARIANT C7 WOTS+C / FORS+C	d=2 h=24
SIGNATURE 3,704 B	
SIGN HASHES 4.3 M	

2^{24} sigs
slow sign

Shrincs stateless
5,777 B
1.7M hashes

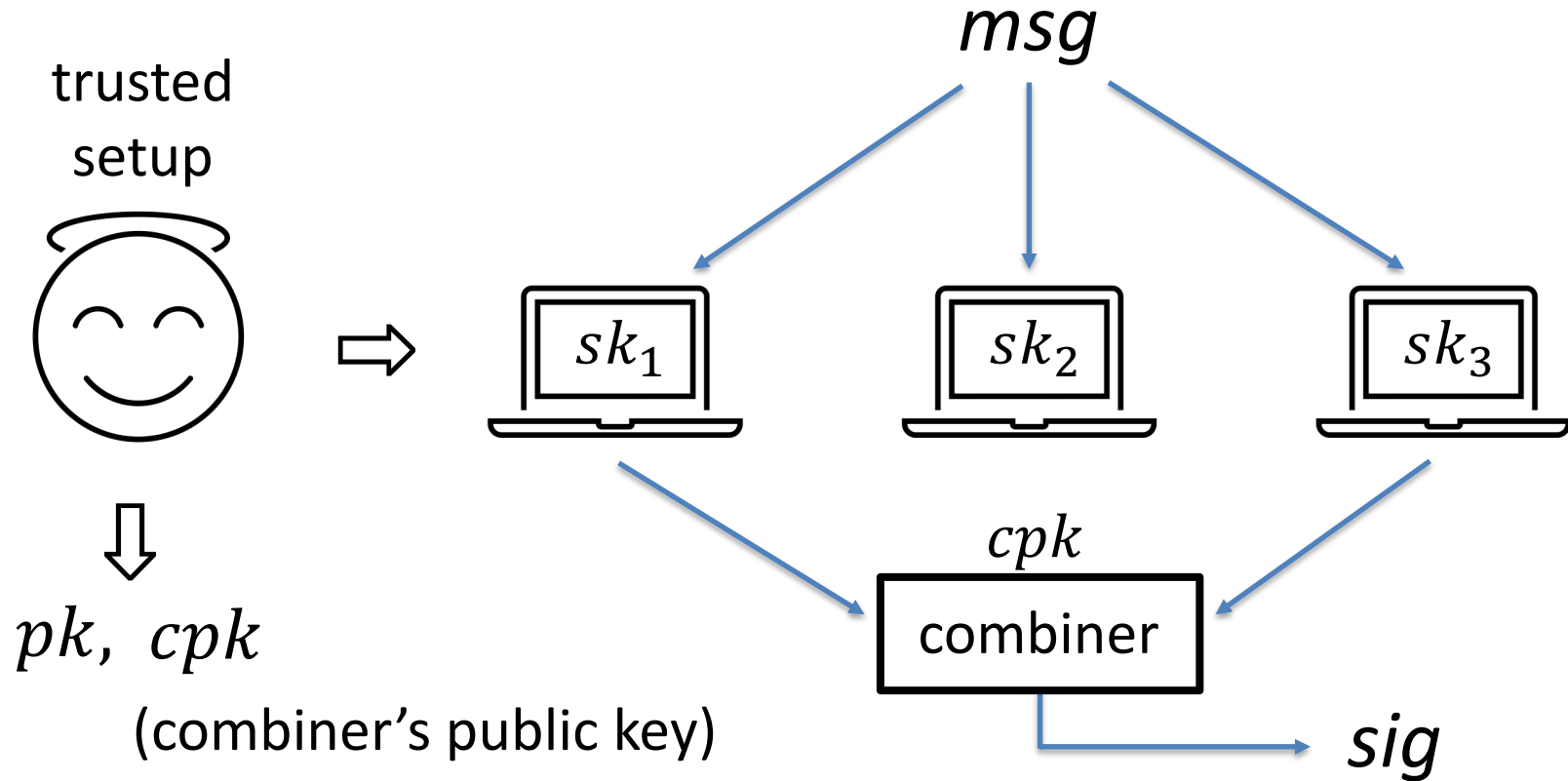
2^{40} sigs

ML-DSA-44: **2,420 B (sig) + 1,312 B (pk)**

ML-DSA-65: **3,309 B (sig) + 1,952 B (pk)**

(defined over a 23-bit prime field)

What about threshold signing for hash-based sigs?



Generic solutions

- **Multi-signing:** $\text{sig} = (t \text{ signatures})$
 $\Rightarrow$ large signatures, exposes t publicly
- **SNARK-based aggregation:** $\text{sig} = \text{SNARK}(t \text{ signatures})$
 $\Rightarrow$ requires on-chain SNARK verification
- **Universal thresholdizer** [[BGG+'18](#)]: $\text{sig} = (\text{same as non-threshold sig})$
 $\Rightarrow$ single round signing, but slow
- **Generic MPC:** $\text{sig} = (\text{same as non-threshold sig}),$ small t .



Our approach: PRAWNS

(with B. Bünz, Justin D., Y. Litchev, and H. Nazirkhanova)

Use lattice crypto to implement the threshold,
... but final signature is a standard hash-based signature

Blockchain is unaware of lattices (or threshold)
... posted sig is a standard (non-threshold) hash-based sig

Main point: thresh. hash-based sigs with large updatable n and t

... but not designed for SLH-DSA (ongoing work)

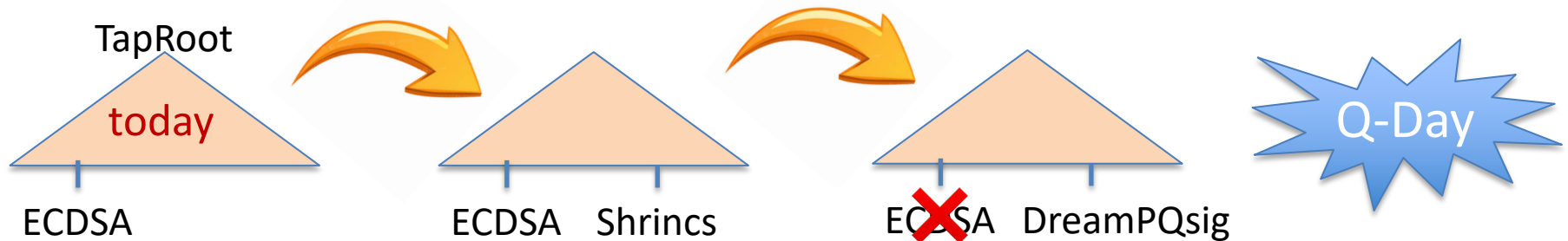
Final thoughts ...

Research on PQ signature is ongoing:

- No reason we can't have 100 bytes sig. + easy threshold + HD

But we need to start the migration now.

A possible two step migration:



Is SHA-256 post quantum secure?

Classical collision attacks on SHA-256 ($N = 2^{256}$):

- Naïve birthday: time = $O(\sqrt{N})$ machine size = $O(\sqrt{N})$
- Rho: time = $O(\sqrt{N})$ machine size = $O(1)$
- Parallel-rho: time = $O(N^{1/6})$ machine size = $O(N^{1/3})$

All these algs. require $\Omega(\sqrt{N})$ evals of the hash function
(a lower bound)

Is SHA-256 post quantum secure?

Quantum collision attacks on SHA-256:

- BHT alg: $\text{time} = O(\sqrt[3]{N})$ $\text{machine size} = O(\sqrt[3]{N})$

Worse than classical parallel-rho, but only $O(\sqrt[3]{N})$ hash evals!

(a lower bound)

Open: quantum collision alg. where

$$\text{time} = O(\sqrt[3]{N}) \qquad \text{machine size} = O(1)$$

⇒ not precluded by any lower bound.

Implication for SHA-256 and Bitcoin

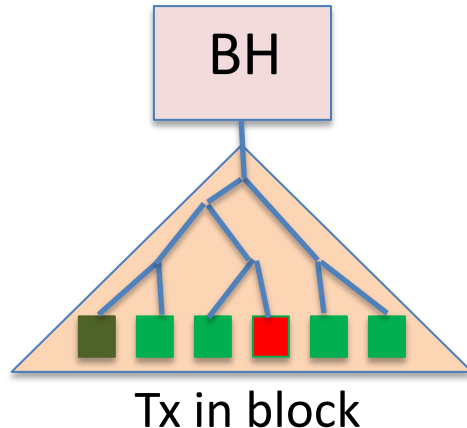
Open: quantum collision alg. where

$$\text{time} = 2^{86}$$

hash evals

machine size = “small”

The red Tx can be opened as one of Tx **after** the block is published



if such collision alg. exists $\Rightarrow$ move to SHA-384
(and parallelizable)

THE END

PRAWNS paper will be posted on eprint in a few days